

Melissa Ning

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EDUCATION

University of Toronto

Toronto, Canada

BASc in Electrical Engineering

Expected May 2029 | PEY Co-op

- **Awards:** NWS Diligence Award (ECE295), Matthew Staikos Memorial Bursary, Dean's Merit Entrance Scholarship

National Tsing Hua University

Hsinchu, Taiwan

Incoming Exchange Student, Semiconductor Engineering with Taiwan Innovations (SETI)

February – June 2027

- **Academic Focus:** Nanotechnology, Microelectronics, and Engineering Business

SKILLS

Programming: C, C++, Python, Assembly

Engineering Tools: Altium Designer, LTspice, MATLAB, Simulink

Hardware & Embedded: PCB Design, Circuit Simulation, Mixed-Signal Design, Power Electronics, Microcontrollers, Oscilloscopes, Vector Network Analyzers, SWD, I2C, UART

EXPERIENCE

Undergraduate Researcher

March 2026 – Present

Intelligent Sensory Microsystems Laboratory, University of Toronto

- **Designed and validated a novel 13.56 MHz transmitter** for nerve-interfacing implantable medical devices, enabling wireless power transfer and data communication through adaptive powering circuitry
- Led end-to-end **hardware and STM32 firmware development**: circuit analysis, simulation, schematic capture, PCB layout, and fabrication
- Performed hardware bring-up, debugging, and validation using **mixed-signal oscilloscopes and VNAs**
- Integrated microcontrollers and sensors into **custom mixed-signal circuits using SWD, I2C, and UART**
- Developed a **Python GUI** for real-time device control and live data monitoring
- Contributed hardware design and results to an **IEEE 2027 conference submission**; currently **co-authoring a paper on programmable near-fields for joint power and sensing**

Power Systems Member

January 2025 – Present

University of Toronto Wind Turbine Student Design Team (UTWind)

- Evaluated power-converter topologies and components for renewable and industrial power applications; defined requirements for efficiency, reliability, voltage regulation, and safe operation
- Designed and simulated **MOSFET-based power converters** for power management under varying loads
- **Developed and debugged PID controller** to regulate output voltage
- Analyzed transient response, efficiency, voltage regulation, and control behavior in **LTspice and MATLAB**

Conference Operational Director

June 2025 – May 2026

Sustainable Engineers Association, University of Toronto

- Led a sustainability-focused engineering conference for **100+ attendees**, managing logistics, budgeting, and scheduling under tight deadlines
- Recruited **10+ industry professionals, academic leaders, and startup representatives** to connect industry and students

PROJECTS

RX Quadrature Receiver | *ECE295: Hardware Design and Communication*

- Designed, simulated, and fabricated a **custom quadrature SDR receiver PCB**, integrating band-pass filtering, quadrature mixing, and low-pass filtering
- Integrated a long-wire antenna, ESP32 local oscillator, PC audio interface, and **Quisk SDR** for end-to-end reception
- Received the **NWS Diligence Award** for outstanding research and technical execution in hardware design

Audio Oscilloscope | *ECE243: Computer Organization*

- Developed an **embedded C application on the DE1-SoC's Nios V processor**, combining 8 kHz stereo audio acquisition through memory-mapped FIFOs with VGA waveform visualization and musical tuning
- Implemented waveform cursors and logarithmic **dBFS conversion** to inspect captured audio levels
- Integrated **PS/2 keyboard controls** for waveform scaling, independent channel pausing, and mode selection